Multilingual Evaluation of Vision–Language Compositionality

Paria Khoshtab
Multimodal Language Understanding Seminar
Saarland University
pkhoshta@mpi-sws.org, pakh00001@stud.uni-saarland.de

Abstract

Vision–language compositionality benchmarks, even those designed for multilingual models, are primarily developed and evaluated in English. It therefore remains unclear whether the abilities and failures in compositional and semantic understanding observed in these benchmarks reflect general model limitations or are influenced by English-specific lexical structure and word order. This work extends SugarCrepe and SugarCrepe++ to Persian, German, and Spanish via machine translation, enabling the compositionality evaluation of multilingual CLIP variants across multiple languages. Our findings reveal performance declines in each translated language, with the largest gaps generally occurring in Persian. Yet the relative difficulty of compositional operations is highly consistent: object swaps are hardest and object replacements easiest across languages and models. These results suggest that language and translation affect absolute performance, while the main compositional failure pattern persists cross-lingually. We release our code at: https://github.com/Theparia/Multilingual-VLM-Compositionality.

1 Introduction

Vision–language models (VLMs) have achieved strong performance on tasks such as image–text retrieval and zero-shot classification, yet their ability to represent compositional meaning remains limited. In particular, a model may recognize the objects and attributes mentioned in a caption while failing to encode how they are related. SugarCrepe addresses this problem using fluent, semantically plausible hard negatives created through object, attribute, and relation replacement, addition, and swapping (Hsieh et al., 2023). Each SugarCrepe example consists of an image, a positive caption that correctly describes it, and a minimally modified negative caption. Its results show that even strong contrastively trained models struggle with compositional binding, especially when the same concepts occur in a different order. SugarCrepe++ extends each example with a second positive caption that is semantically equivalent to the first but lexically different (Dumpala et al., 2024). This format allows the benchmark to measure whether models prefer semantic equivalence over superficial lexical similarity, both with and without access to the image.

Since both benchmarks are English-only, it remains underexplored whether their reported failures reflect general limitations in compositional representation or English-specific lexical and syntactic properties. This is particularly relevant for the swap subsets, as some languages have different word orders. For example, Persian commonly follows a subject–object–verb order, unlike the predominantly subject–verb–object order of English and Spanish.

We therefore investigate the following research question: *How does evaluation language affect the compositional and semantic-understanding performance of multilingual VLMs on SugarCrepe and SugarCrepe++?* We focus on cross-lingual performance gaps and whether the relative difficulty of the compositional operations is preserved across languages.

We answer this by machine-translating both benchmarks into German, Spanish, and Persian and evaluating two multilingual OpenCLIP models (Ilharco et al., 2021), with XLM-R base and large text encoders, on the resulting sets. Our results show consistent degradation in all three target languages, largest for Persian on SugarCrepe, but an unchanged ordering of subset difficulty across all languages and both models: swapped objects remain hardest and replaced objects easiest everywhere. These results indicate that evaluation language affects model performance, but does not fully explain the observed limitations of VLMs in compositional and semantic understanding.

2 Related Work

Vision-language compositionality. A substantial line of work shows that contrastively trained VLMs such as CLIP behave like "bags-of-words," encoding the objects mentioned in a caption while largely ignoring attributes, relations, and word order (Yuksekgonul et al., 2023). Thrush et al. (2022) introduced Winoground, whose minimal pairs share an identical set of words in a different order and on which strong VLMs fall below the random-chance group baseline. Related benchmarks such as ARO (Yuksekgonul et al., 2023), CREPE (Ma et al., 2023), and VALSE (Parcalabescu et al., 2022) scaled this diagnosis across relations, order, and targeted linguistic phenomena. Hsieh et al. (2023) showed that such rule-based benchmarks are "hackable," with blind text-only models exploiting fluency and plausibility artifacts to outperform CLIP. They proposed SugarCrepe, which uses fluent and adversarially refined hard negatives based on REPLACE, SWAP, and ADD operations. SugarCrepe++ (Dumpala et al., 2024) adds lexically distinct but semantically equivalent positives to separate semantic from surface sensitivity, and BiVLC (Miranda et al., 2024) contributes hard-negative images for bidirectional retrieval. Crucially, all of these benchmarks are constructed and evaluated in English, leaving open whether the failures they report are properties of the models or language.

Multilingual VLMs and evaluation. Multilingual CLIP variants replace or distill the text encoder with XLM-R, either through teacher learning (Carlsson et al., 2022; Chen et al., 2023) or by contrastive training on LAION-5B corpus (Schuhmann et al., 2022; Cherti et al., 2023), and the models we study belong to this family. Evaluation of such models, however, has centered on classification, retrieval, VQA, and NLI, as in IGLUE (Bugliarello et al., 2022) and Babel-ImageNet (Geigle et al., 2024); the latter reports a significant gap between English and other languages but does not isolate compositional understanding. Gomes et al. (2025) machine-translates VALSE into several European languages to validate multilingual CLIPScore as an image-captioning evaluation metric and finds that multilingual models retain high cross-lingual correlation with human judgments. In contrast, we investigate how the SugarCrepe-style compositional and semantic understanding of multilingual VLMs varies across evaluation languages.

P_1 **(positive)**

یک سگ که پشت یک جفت کفش مشکی نشسته است.

EN: A dog sitting behind a pair of black shoes.

P_2 **(equivalent positive)**

یک جفت کفش سیاه در مقابل یک سگ قرار دارد.

EN: A pair of black shoes is positioned in front of a dog.

N **(hard negative)**

یک جفت کفش سیاه که پشت یک سگ قرار دارند.

EN: A pair of black shoes sitting behind a dog.

Figure 1: A Persian `swap_obj` example with its English source captions. P_1 and N form the SugarCrepe positive–negative pair by reversing the positions of the dog and shoes; P_2 is the lexically different, semantically equivalent positive added by SugarCrepe++.

3 Experimental Setup

3.1 Datasets

SugarCrepe contains 7,511 examples across seven subsets: 2,062 `add_obj`, 692 `add_att`, 1,652 `replace_obj`, 788 `replace_att`, 1,406 `replace_rel`, 245 `swap_obj`, and 666 `swap_att`. SugarCrepe++ contains 4,757 items: the same five replace and swap subsets, excluding `add_obj` and `add_att` because their negatives may require the image to establish semantic inequivalence. Both datasets use COCO 2017 validation images (Lin et al., 2014) and are evaluation-only.

To enable cross-lingual evaluation, we construct aligned multilingual versions of both benchmarks in German (`de-DE`), Spanish (`es-ES`), and Persian (`fa-IR`). For every English example, we retain the original sample ID, image filename, and source caption(s) and add translated fields for each target language. In SugarCrepe, we translate `caption` and `negative_caption`; in SugarCrepe++, we additionally translate `caption2`. Figure 1 shows this structure with an aligned Persian `swap_obj` example and its English source captions. Translations are generated with `google/translategemma-4b-it` (Finkelstein et al., 2026) through the Hugging Face Transformers library. We use deterministic decoding and process records in resumable batches. Captions belonging to the same example are generated within the same batch, but each caption is independently prompted, so the model is not explicitly informed of the positive–negative relationship. This design is important because independent translation can remove the minimal lexical contrast that defines a hard negative. After translation, we thus check for exact collisions where captions become identical after normalization. The colliding examples are excluded from our results in the following sections. The number of translation collisions by language and subset is reported in Appendix A.

3.2 Models

We evaluate the two multilingual dual-encoder models from the original SugarCrepe paper, available through OpenCLIP, that support all target languages in our benchmark: `xlm-roberta-base-ViT-B-32` and `xlm-roberta-large-ViT-H-14`. Both models are trained on the large-scale `LAION-5B` (Schuhmann et al., 2022) image-text dataset and combine an XLM-R text encoder (Conneau et al., 2020) with a CLIP-style contrastive image–text architecture (Radford et al., 2021).

3.3 Evaluation Procedure

We follow the original evaluation protocols of SugarCrepe and SugarCrepe++. For SugarCrepe, a sample is counted as correct if the image is more similar to the positive caption than to the negative caption. Given cosine similarity $s(\cdot, \cdot)$, a hit is defined as

$$\mathbf{1}[s(I, P) > s(I, N)]\,, \tag{1}$$

where I denotes the image, P the positive caption, and N the negative caption. Ties are counted as incorrect.

For SugarCrepe++, we report both Image-to-Text (ITT) and Text-Only (TOT) accuracy. ITT requires both positive captions to be more similar to the image than the negative caption:

$$\begin{aligned} \mathrm{ITT} = \mathbf{1}\big[&s(I, P_1) > s(I, N) \\ &\wedge s(I, P_2) > s(I, N)\big], \end{aligned} \tag{2}$$

where P_1 and P_2 are the semantically equivalent positive captions. TOT evaluates whether the textual representations capture their semantic equivalence without access to the image:

$$\begin{aligned} \mathrm{TOT} = \mathbf{1}\big[&s(P_1, P_2) > s(P_1, N) \\ &\wedge s(P_1, P_2) > s(P_2, N)\big]. \end{aligned} \tag{3}$$

For all metrics, we report accuracy for each subset, as well as the unweighted macro-average across subsets for each language.

3.4 Implementation Details

We implement translation and evaluation in Python using PyTorch, Hugging Face Transformers, and OpenCLIP, with all experiments conducted on a single NVIDIA Tesla V100 GPU (32 GB) under Debian GNU/Linux 13. TranslateGemma-4B is loaded in bfloat16 precision without 8-bit quantization. Translation uses deterministic decoding with `do_sample=False`, `num_beams=1`, and `max_new_tokens=128`. Unless otherwise specified, all other hyperparameters are kept at their default values. For evaluation, images are converted to RGB and processed using each checkpoint's OpenCLIP transformation, while captions are encoded with its corresponding tokenizer. Image and text embeddings are L2-normalized, and their dot product is used as cosine similarity. We apply the same preprocessing, scoring rules, and model checkpoints to every language, without fine-tuning or language-specific calibration.

4 Results

We present several key findings from our evaluation, with results reported in Table 1 for SugarCrepe and Table 2 for SugarCrepe++. Across both benchmarks and models, compositional weaknesses persist across all evaluated languages, with similar relative difficulty patterns despite their syntactic differences. Absolute performance, however, degrades under translation, most for Persian. We support each claim in turn below.

Cross-lingual performance gap. The aggregate results show the same picture for both benchmarks: every target language falls below English for both models. On SugarCrepe (Table 1) the macro-average drops from 78.12 (EN) to 73.91 (DE), 74.25 (ES), and 70.99 (FA) for Base, and from 81.57 to 77.71, 77.22, and 75.07 for Large. SugarCrepe++ (Table 2) reproduces this under a stricter protocol: Base ITT falls from 63.74 (EN) to 53.26 (FA), and Large ITT from 65.26 to 58.92 (FA). German and Spanish cluster together and stay closer to English, while Persian is the largest gap in almost every setting, giving a stable EN $>$ DE $\approx$ ES $>$ FA ranking. The relative drop is also sharper on SugarCrepe++: the Base ITT English–Persian gap of 10.48 points is a 16.4% relative decline, versus 9.1% on SugarCrepe, which may reflect the stricter requirement that both lexically distinct positives outperform the hard negative. Therefore, compositional performance is not invariant to the evaluation

			SWAP		REPLACE			ADD		
Source	Model	Lang.	Object	Attribute	Object	Attribute	Relation	Object	Attribute	Average
LAION	XLM-R Base–ViT-B/32	EN	63.27	67.62	93.30	83.91	69.70	87.77	81.25	78.12
		DE	55.51	67.02	91.95	79.44	65.70	85.20	72.53	73.91
		ES	55.10	70.33	91.58	79.82	64.44	83.50	75.00	74.25
		FA	54.29	65.36	86.98	78.54	61.48	79.52	70.78	70.99
	XLM-R Large–ViT-H/14	EN	63.67	72.14	96.99	86.08	72.67	93.11	86.34	81.57
		DE	58.78	67.92	95.58	83.40	69.63	91.51	77.18	77.71
		ES	56.73	70.18	95.21	83.40	68.15	89.08	77.76	77.22
		FA	58.78	68.83	91.77	79.18	67.48	85.64	73.84	75.07

Table 1: Performance of the multilingual OpenCLIP models on SugarCrepe after excluding the union of 92 collapsed records from all languages. Results are reported as accuracy (%). Average denotes the mean accuracy across the seven subsets, with each subset weighted equally.

			Swap Object		Swap Attribute		Replace Object		Replace Attribute		Replace Relation		Average	
Source	Model	Lang.	ITT	TOT	ITT	TOT	ITT	TOT	ITT	TOT	ITT	TOT	ITT	TOT
LAION	XLM-R Base–ViT-B/32	EN	44.77	28.03	55.28	54.33	89.35	95.60	73.42	81.05	55.90	57.76	63.74	63.35
		DE	41.00	34.31	52.60	55.28	86.99	90.43	68.42	70.79	51.09	52.80	60.02	60.72
		ES	36.82	34.31	50.08	51.18	85.20	89.48	63.55	68.82	47.13	48.99	56.56	58.56
		FA	35.56	35.15	42.83	51.34	80.68	83.04	61.58	64.21	45.65	44.57	53.26	55.66
	XLM-R Large–ViT-H/14	EN	43.10	28.45	54.17	51.50	93.88	94.45	76.05	79.34	59.08	60.64	65.26	62.88
		DE	38.49	33.05	54.65	58.74	91.45	87.05	70.00	65.79	53.34	52.33	61.59	59.39
		ES	37.24	35.15	49.92	49.61	89.99	86.48	66.05	64.34	51.24	49.69	58.89	57.05
		FA	38.49	30.54	51.02	53.39	86.99	82.27	65.00	66.05	53.11	49.38	58.92	56.33

Table 2: Performance of the multilingual OpenCLIP models on SugarCrepe++ after excluding the union of 267 collapsed records from all languages. Results are reported as accuracy (%). Average denotes the mean accuracy across the five subsets, with each subset weighted equally.

language, with the largest degradation observed for Persian, the lower-resource and typologically more distant language in our study.

The difficulty ordering is language-invariant. Despite these absolute gaps, the relative difficulty of the operations is preserved. Across all eight rows of Table 1 and under both metrics of Table 2, `swap_obj` is the hardest subset and `replace_obj` the easiest, without exception. This coarse structure with attribute operations in between transfers cleanly to Persian, whose subject–object–verb order differs from the subject–verb–object order of English and Spanish. The only systematic change here is that in almost all three target languages `replace_rel` overtakes `swap_att` as the second-hardest SugarCrepe subset (e.g., FA Base: 61.48 vs. 65.36), whereas in English the two swaps are the two hardest. Relations and attribute binding are thus the operations most sensitive to translation, consistent with the difficulty of preserving fine-grained modifier and relational structure across languages; attribute addition shows this too, with the largest per-subset drop into Persian (-10.47 for Base, -12.50 for Large).

English is not advantaged on object swaps. The text-only (TOT) `swap_obj` results provide particularly clear evidence that this difficulty is not specific to English. All three translated languages outperform English for both models (Base: 34.31/34.31/35.15 for DE/ES/FA vs. 28.03 for EN; Large: 33.05/35.15/30.54 vs. 28.45). This shows that changing the evaluation language does not remove the models' difficulty in distinguishing captions in which the same objects exchange roles. Since English is actually the lowest-performing language in this setting, the weakness cannot be explained solely by English-specific word order. These results therefore suggest that the weakness on compositional swaps is model-intrinsic rather than a property of the source language.

Effect of model scale. Scaling from XLM-R Base–ViT-B/32 to XLM-R Large–ViT-H/14 improves macro-averages of SugarCrepe in every language and improves SugarCrepe++ ITT throughout, with the largest gain for Persian ($53.26 \rightarrow 58.92$). However, it does not help the text-only setting: SugarCrepe++ TOT is comparable to or slightly below Base for English, German, and Spanish (e.g., EN

$63.35 \rightarrow 62.88$) and improves only marginally for Persian. Scale therefore strengthens image-grounded discrimination but leaves the text encoder's ability to recognise semantic equivalence largely unchanged, and it narrows but does not close the cross-lingual gap.

Qualitative analysis. The Persian `swap_obj` example in Figure 1 illustrates why this subset is uniformly hardest. The positive P_1, the equivalent positive P_2, and the hard negative N share essentially the same content words (dog, black shoes, behind/in front of) and differ only in which object occupies which slot of the spatial relation. A model that encodes the lexical content but not the binding between objects and their relational roles cannot separate P_1 from N, and this holds equally in the Persian rendering, where the reordering required by subject–object–verb structure does not change the outcome. Concretely, XLM-R Base prefers P_1 over N in English but prefers the negative in Persian, whereas XLM-R Large resolves the item in both languages. Under the TOT metric, however, both models fail it in both languages, ranking N closer to each positive than the two positives are to each other. The same words in a different arrangement thus remain the failure mode in every language. The similarity scores for this example are detailed in Appendix B.

5 Conclusion

Compositionality benchmarks for vision–language models are mostly English-only, leaving open whether the failures they report for English are properties of the models or artifacts of English lexical structure and word order. We addressed this by machine-translating SugarCrepe and SugarCrepe++ into German, Spanish, and Persian and evaluating two multilingual OpenCLIP models under identical preprocessing and scoring in all four languages. Our central finding is twofold. Absolute performance degrades consistently under translation, most for Persian, so measured compositional ability depends in part on the evaluation language and its resource level. The relative difficulty of the compositional operations, however, is preserved across languages: object swaps remain hardest and object replacement easiest everywhere, and on text-only object swaps, English is the weakest case.

Our study also has limitations that suggest directions for future work. Because our multilingual sets are produced by machine translation, some cross-lingual degradation may stem from translation artifacts rather than model behavior; human-verified or natively authored evaluation data would disentangle the two. Extending the analysis to lower-resource and typologically more diverse languages, and to multilingual VLMs beyond the XLM-R OpenCLIP family, would test how far these conclusions generalize. Finally, pairing our translated captions with hard-negative images, as in BiVLC (Miranda et al., 2024), would allow the same cross-lingual questions to be asked in the image-to-image direction.

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A Translation Collision Analysis

Table 3 reports exact collisions after Unicode NFKC normalization, whitespace collapsing, trimming, and case folding. A SugarCrepe record collides when its positive and negative captions become identical. A SugarCrepe++ record is counted once when any two of P_1, P_2, and N become identical, even if multiple pairs collide.

Benchmark	Subset	DE	ES	FA	Union
SugarCrepe	`add_obj`	0	0	1	1
	`add_att`	2	1	1	4
	`replace_obj`	7	7	12	24
	`replace_att`	2	3	1	5
	`replace_rel`	25	19	24	56
	`swap_obj`	0	0	0	0
	`swap_att`	2	0	0	2
	Total	**38**	**30**	**39**	**92**
SugarCrepe++	`replace_obj`	32	31	40	84
	`replace_att`	8	14	13	28
	`replace_rel`	52	55	46	118
	`swap_obj`	4	4	2	6
	`swap_att`	21	10	10	31
	Total	**117**	**114**	**111**	**267**

Table 3: Numbers of translated records containing an exact caption collision. Language columns count unique affected records in each translation. Union deduplicates records affected in multiple languages.

Collisions are more frequent in SugarCrepe++ because it contains three captions and therefore more opportunities for pairwise collision.

B Similarity Scores for the Qualitative Example

Tables 4 and 5 report the raw cosine similarities and per-metric outcomes for the `swap_obj` record shown in Figure 1 (record 77, image `000000412240.jpg`). The image–caption similarities are shared between the two benchmarks, so the SugarCrepe columns $s(I,P)$ and $s(I,N)$ are identical to the SugarCrepe++ columns $s(I,P_1)$ and $s(I,N)$.

C Disclaimer: Use of AI

We utilized AI assistants (Claude Sonnet 5 and ChatGPT) for grammatical editing, stylistic polishing of the manuscript, and assistance with code development. All research decisions, experimental design, analysis, and interpretation of the results were carried out and verified by the author.

Model	Lang.	$s(I,P)$	$s(I,N)$	Margin	Correct
Base	EN	0.2488	0.2365	$+0.0123$	✓
	DE	0.2346	0.2430	-0.0084	
	ES	0.2257	0.2281	-0.0025	
	FA	0.2216	0.2300	-0.0084	
Large	EN	0.2390	0.2297	$+0.0093$	✓
	DE	0.2438	0.2203	$+0.0236$	✓
	ES	0.2277	0.2067	$+0.0209$	✓
	FA	0.2443	0.2320	$+0.0123$	✓

Table 4: SugarCrepe outcome for the record in Figure 1. Margin is $s(I,P) - s(I,N)$; a record is correct when the margin is positive. A checkmark denotes a correct outcome; blank entries denote incorrect outcomes.

		Image-to-text				Text-only			
Model	Lang.	$s(I,P_1)$	$s(I,P_2)$	$s(I,N)$	ITT	$s(P_1,P_2)$	$s(P_1,N)$	$s(P_2,N)$	TOT
Base	EN	0.2488	0.2330	0.2365		0.9456	0.9673	0.9756	
	DE	0.2346	0.2234	0.2430		0.9380	0.9364	0.9587	
	ES	0.2257	0.2119	0.2281		0.9271	0.9425	0.9639	
	FA	0.2216	0.2057	0.2300		0.9117	0.9302	0.9691	
Large	EN	0.2390	0.2358	0.2297	✓	0.9265	0.9604	0.9534	
	DE	0.2438	0.2163	0.2203		0.9052	0.9104	0.9485	
	ES	0.2277	0.2100	0.2067	✓	0.9266	0.9257	0.9635	
	FA	0.2443	0.2270	0.2320		0.8604	0.9135	0.9534	

Table 5: SugarCrepe++ outcome for the record in Figure 1. ITT and TOT are marked correct according to the conjunction criteria defined in Section 3. A checkmark denotes a correct outcome; blank entries denote incorrect outcomes.